

Math 4101 Mid-term 1
Version C

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Time: 50 minutes

Total: 100 points

Directions. Write complete proofs or arguments. Clearly state any definitions you use. Unless otherwise stated, all numbers are real. Throughout this exam, $\mathbb{N} = \{1, 2, 3, \dots\}$.

Problem 1 (30 points).

Let

$$S = \{x \in \mathbb{R} : x^2 \in \mathbb{Q}\}.$$

- (a) (15 points) Prove that S is countably infinite.
- (b) (15 points) Prove that $\mathbb{R} \setminus S$ is uncountable.

Proof. The set $\mathbb{Q}$ is countably infinite, done through zigzag argument. Since $\mathbb{Q} \subset S$, S inherits the infinite property. To prove countability, let $S = \{\pm\sqrt{\alpha} : \alpha \in \mathbb{Q}, \alpha \geq 0\}$. If for some $\pm\sqrt{\alpha} \in \mathbb{Q}$, we can simply pair it with its corresponding element within $\mathbb{Q}$. Denote this as S' , which is countably infinite. We do not need to prove surjectivity or injectivity for this, since S' is simply $\mathbb{Q}$. Let $S'' := \{\pm\sqrt{\alpha} : \sqrt{\alpha} \notin \mathbb{Q}, \alpha > 0, \alpha \in \mathbb{Q}\}$, which is a subset of S . Define a function $f : \{\alpha \in \mathbb{Q}_{>0} : \sqrt{\alpha} \notin \mathbb{Q}\} \times \{+, -\} \rightarrow S''$. Let $f(\alpha, +) = \sqrt{\alpha}$ and $f(\alpha, -) = -\sqrt{\alpha}$. To prove injectivity, suppose $f(\alpha, \pm) = f(\beta, \pm)$. Thus, $\pm\sqrt{\alpha} = \pm\sqrt{\beta}$. Squaring both sides, we get $\alpha = \beta$. Now, given this, the signs matter, since $\sqrt{\alpha} > 0 \implies \alpha > 0$, thus $-\sqrt{\alpha} \neq \sqrt{\alpha}$. Thus, the signs must match in order to have equivalent values, proving injectivity.

For surjectivity, take $y \in S''$. Thus, $y = \pm\sqrt{\alpha}$ for some $\alpha \in \mathbb{Q}_{>0}$. Set $f(\alpha, +)$ if $y = \sqrt{\alpha}$, and $f(\alpha, -)$ if $y = -\sqrt{\alpha}$. Either way, y is being hit. Therefore, since f is both injective and surjective, it is a bijection of a countably infinite set ($\mathbb{Q}_{>0} \subset \mathbb{Q}$ is infinite).

Thus, since S' and S'' are countably infinite, $S = S' \cup S''$ is countably infinite, since a finite union of countable sets is countable.

Suppose $\mathbb{R} \setminus S$ is countable. Suppose $\mathbb{R} = (\mathbb{R} \setminus S) \cup S$. Since $\mathbb{R} \setminus S$ is countable, and S is also countable, the union of the two should also be countable. But we know that $\mathbb{R}$ is uncountably infinite, due to Cantor's diagonal argument. Thus, this is a contradiction. Therefore, $\mathbb{R} \setminus S$ is uncountable. $\square$

Problem 2 (30 points).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and write $s = \sup A$.

- (a) (15 points) Prove that for every $\varepsilon > 0$, there exists $a \in A$ such that

$$s - \varepsilon < a \leq s.$$

- (b) (15 points) Define

$$T = \left\{ a - \frac{1}{n} : a \in A, n \in \mathbb{N} \right\}.$$

Prove that $\sup T = s$. You may use part (a), even if you have not proved it.

Proof. We are given that $s = \sup A$. Therefore, $\forall a \in A, a \leq s$. We are told to prove that there exists an element $a \in A$ such that $s - \epsilon < a \leq s$. Suppose not, and that $\forall a \in A, a \notin (s - \epsilon, s]$. Therefore, $a \leq s - \epsilon$. It's given that s is the least upper bound, and we found another upper bound that is less than s , so this is contradictory.

Let $T := \{a - \frac{1}{n} : a \in A, n \in \mathbb{N}\}$. Let $t \in T$. Thus, we see $t = a - \frac{1}{n}$ for some $a \in A, n \in \mathbb{N}$. We may deduce that $t = a - \frac{1}{n} < a \leq s$. Thus, s is an upper bound. Since $\sup A = s$, from the last proof, $s - \epsilon < t \leq s$. This is proven because $\exists a > s - \epsilon/2$ and $\exists \frac{1}{n} < \epsilon/2$ such that $t > s - \epsilon$. Since ϵ is arbitrary, s is the least upper bound for T . Thus, $\sup T = s$. $\square$

Problem 3 (40 points).

Use the standard metric $d(x, y) = |x - y|$ on $\mathbb{R}$.

- (a) (20 points) Let $U, V \subseteq \mathbb{R}$ be open. Prove directly from the definition of openness that $U \cap V$ is open.
- (b) (20 points) Suppose $U_n \subseteq \mathbb{R}$ is open for every $n \in \mathbb{N}$. Must

$$\bigcap_{n=1}^{\infty} U_n$$

be open? Give a proof or an explicit counterexample. If you give a counterexample, determine the intersection and justify why it is not open.

Proof. Let $X = U \cap V$. If $X = \emptyset$, then it is open by definition. Thus, let $X \neq \emptyset$. Let $x \in X$. Since $x \in U$, $\exists r_1 > 0$ such that $B_{r_1}(x) \subseteq U$. Likewise, $x \in V \implies \exists r_2 > 0$ such that $B_{r_2}(x) \subseteq V$. Now let $r = \min(r_1, r_2)$. Since $r \leq r_1$, $B_r(x) \subseteq B_{r_1}(x) \subseteq U$. Likewise, $r \leq r_2 \implies B_r(x) \subseteq B_{r_2}(x) \subseteq V$. Thus, $B_r(x) \subseteq U \cap V$. $\square$

Proof. Let $X = \bigcap_{n=1}^{\infty} U_n$. Assume $X \neq \emptyset$. Let $x \in X$. Since U_n is open for all n , then $B_{r_n}(x)$ for each n . Now, let $r = \min(r_1, r_2, r_3, \dots) > 0$. But the minimum for infinitely many positive integers doesn't need to be positive. It can be zero. Thus, X isn't open. $\square$